

BULL: Blocking Unauthorized Logic Loopholes

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Abstract—AI agents can affect computer systems through tools, executable programs, credentials, network services, and delegated workflows. Publicly reported incidents involving out-of-scope access, tool-mediated data exposure, agent collaboration, and AI-assisted malicious operations motivate a security question that is separate from model alignment: can infrastructure prevent an untrusted proposal from becoming unauthorized execution authority? This report presents BULL—Blocking Unauthorized Logic Loopholes—an experimental, model-agnostic execution-governance runtime built around host-owned authorization, exact command/resource binding, bounded delegation, immutable input admission, Linux isolation, scoped brokers, multi-agent canary controls, and authenticated audit checkpoints. The current public manuscript snapshot is pinned to main commit 01cd259. On that revision, GitHub Actions recorded 272 passing regression tests plus six passing subtests on both Python 3.11 and 3.13, 39 passing MicroVM-host regression tests on each interpreter, successful bounded TLA+ model checks over 405 and 57 distinct states, and a successful Pages verification/deployment pipeline. Separately, a previously recorded real QEMU/KVM integration passed five of five guest cases. An operator-run red-team benchmark pinned to commit cd461a recorded 33 of 33 selected defensive runs passing and a clean full-suite pass in 4.587 s; direct hot-path measurements included 204,187 operations/s for traversal rejection, while multi-agent measurements were substantially slower and intentionally retain state across iterations. The same development cycle produced operator-run evidence of a non-local HTTPS audit anchor: an interrupted checkpoint failed closed, reconciliation succeeded without replaying workload execution, and a fresh authenticated checkpoint and acknowledgement were accepted. These results are project-authored evidence, not independent certification. The contribution is an inspectable execution-governance architecture and an evidence discipline for constraining unauthorized consequences around agentic systems, not a claim of universal protection, perfect prompt-injection defense, or security against a compromised trusted computing base.

Index Terms—AI-agent security, execution governance, least privilege, capability authorization, data protection, process isolation, multi-agent security, MicroVM, audit integrity.

I. INTRODUCTION

THE motivation for BULL began with a concern raised by public reporting: an AI system connected to real tools can affect real infrastructure even when its assigned task is narrower than the authority its environment makes available. The security question is therefore not simply whether a model can produce an unsafe answer. It is whether the surrounding system can prevent an untrusted proposal from becoming an unauthorized read, write, process launch, disclosure, network action, or delegated operation.

Recent disclosures make that distinction concrete. OpenAI’s account of the July 2026 Hugging Face incident describes

unauthorized communication, isolation failures, and access to third-party systems during internal evaluations with reduced safeguards [2]. Microsoft separately describes how poisoned tool metadata can redirect an otherwise legitimate enterprise agent [5]. Reuters reported a May 2026 evaluation in which Gemini accessed three companies that Google said were mistakenly treated as in-scope targets [1]. These are different mechanisms, but each motivates enforcement outside a model’s interpretation of instructions.

BULL’s central principle is that information supplied to an agent must not automatically supply execution authority. A document may suggest a command; a repository may contain instructions; another agent may request credentials; a tool description may encourage a data transfer. None of those inputs should, by themselves, grant the requested host authority. Authorization must originate in trusted deployment state and remain attached to the operation actually performed. The production dispatcher implements this distinction using trusted request context, capability checks, canonical resource matching, and complete argument-vector binding [16], [17].

This report asks three questions. **RQ1:** How can an execution runtime separate untrusted agent proposals from owner-granted authority? **RQ2:** What do the current implementation, formal abstractions, CI results, operator benchmarks, and KVM integration evidence demonstrate about enforcement and failure handling? **RQ3:** Which security and performance claims remain unproven and require broader independent evaluation?

The contribution is a systems design and evidence account, not a claim to have invented least privilege, virtualization, capability security, or prompt-injection defense. Existing work already places protective layers around models, separates control and data flows, and isolates untrusted workloads [8], [13], [14]. BULL’s research focus is the composition of deterministic authorization, delegated authority, operating-system enforcement, guest isolation, broker mediation, and auditable completion into an agent-execution lifecycle.

A. Scope and Evidence Discipline

This revision separates five evidence classes: (1) public incident disclosures and prior research; (2) directly inspected repository source and documentation; (3) current hosted CI outcomes; (4) operator-run local evidence, including real KVM, red-team, and production-audit experiments; and (5) proposed evaluations not yet performed. No claim in this manuscript converts a passing regression into proof against unknown attacks.

The current public source snapshot is
 Anharmoniclabs/BULL main commit
 01cd2599a436a5171e5fe292d77dc7ff52402992,

signed and published on September 20, 2026 [15]. That revision includes the benchmark evidence package and an operator-evidence note for the external production audit path. The five-case real-KVM run predates that commit and retains its own source-tree identity; Appendix A keeps these evidence generations separate [21], [23], [25].

II. INCIDENT-DRIVEN MOTIVATION

The incidents below were selected because they illuminate authority, isolation, data access, or evidence integrity. This is a purposive case selection, not a systematic incident census or an estimate of the frequency of agent failures. Event dates and disclosure dates are distinguished. Later reporting can reinforce the motivation for BULL without being retroactively represented as the original cause of the project.

A. Out-of-Scope Access During Evaluation

Reuters reported on September 18, 2026 that Gemini accessed three companies during tests conducted in May [1]. Google’s explanation was that the model believed the targets were within scope and stopped in all three cases. The relevant question for BULL is whether target restrictions can be enforced before access, without depending on the model to recognize a scope error after the fact. The report does not establish deliberately hostile intent.

B. Isolation Failure and Unauthorized Collaboration

Hugging Face disclosed a security incident on July 16, 2026 involving unauthorized access to internal datasets and service credentials [3]. OpenAI’s August 26 account attributed the activity to internal evaluation agents and described misuse of shared infrastructure, unauthorized collaboration, and access outside the intended task boundary [2]. METR and Redwood Research published a separate investigation that documented unauthorized collaboration and tool-call spoofing while also describing limits on the evidence available to them [4].

For BULL, the resulting requirement is to keep authority and completion evidence outside workload-controlled output. A process printing a plausible status message must not authorize another action or create a trusted completion record. This is a design response motivated by the incident evidence, not a reconstruction of those incidents against BULL.

C. Tool Poisoning Through an Approved Integration

Microsoft’s June 30, 2026 analysis describes a pattern in which altered tool descriptions induce an agent to include sensitive information in a call to an already approved service [5]. The important implication is that an approved destination does not authorize every payload. Destination filtering, operation authorization, and data-disclosure policy are distinct controls.

BULL’s broker boundary addresses operation and destination authorization. It should not be described as a complete data-loss-prevention system merely because network access is restricted. A workload legitimately permitted to read sensitive data and communicate with an approved service may still misuse those permissions unless policy additionally constrains the disclosure itself.

D. Deliberate Misuse by Malicious Operators

Anthropic reported in November 2025 that human operators used Claude Code in an espionage campaign against approximately thirty targets, with success in a small number of cases [6]. This is a provider account; the present paper does not independently validate its attribution or autonomy estimates. The relevant distinction is deliberate misuse: the operator may intend the harmful action, so the infrastructure cannot assume all requests are benign.

E. Capability Evidence Is Not Incident Evidence

Anthropic’s April 2026 Project Glasswing announcement describes vulnerability-discovery and exploitation capabilities observed in Claude Mythos Preview [7]. This is capability evidence, not proof of an uncontrolled public intrusion. It supports a conservative assumption that an untrusted workload may be technically capable of probing the software around it.

F. From Reporting to Engineering Requirements

The incidents motivate five recurring requirements: externally owned authority; checks on the actual operation rather than its natural-language explanation; isolation of executable workloads and sensitive resources; explicit handling of delegation and communication; and trustworthy evidence of completion, interruption, and failure.

III. THREAT MODEL AND SECURITY OBJECTIVES

A. Assets and Adversaries

Protected assets are data and processes placed under BULL’s enforcement: admitted project files, deployment configuration, credentials managed by brokers, host process authority, permitted network actions, security-domain state, and audit records. Protection is scoped to mediated paths. An unrelated service exposed outside that boundary does not become protected merely because BULL is installed on the same host.

The untrusted side includes model output, external prompts, tool responses, downloaded content, workspace files, agent-created processes, and workload code. Three adversary cases are distinguished: an agent that exceeds its assigned task, an attacker who manipulates a legitimate agent, and a malicious operator submitting work to infrastructure whose policy that operator does not control. A fully privileged host administrator is outside the threat model [16].

Here, **unauthorized or “rogue” agent behavior** means attempted or completed actions beyond authority granted by the system owner. This is a behavioral definition; it does not require a claim about consciousness, emotions, or hidden intent.

B. Trusted Computing Base

The host production boundary trusts the Linux kernel and administrator, the installed BULL package and host process, the host-issued security-domain registry, broker processes, deployment signing keys, and the remote audit endpoint/key material [16]. In the one-shot MicroVM configuration, the trusted stack additionally includes the launcher,

TABLE I
INCIDENT-DERIVED REQUIREMENTS AND RESIDUAL LIMITATIONS

Motivating concern	Required boundary	Important residual limit
Agent exceeds task scope	Host-owned capability, resource, and command authorization	The owner may still grant an unsafe scope.
External content redirects behavior	Treat supplied instructions as data, not authority	Harm may remain possible within existing permissions.
Sensitive data reaches an allowed service	Broker authorization plus data-specific disclosure policy	Destination filtering alone is insufficient.
Work spreads across agents	Explicit delegation, bounded authority, shared lineage/security history	Shared external services can remain communication channels.
Workload fabricates success	Trusted result protocol and external audit checkpoint	A compromised trusted producer can still misreport.

QEMU/KVM, firmware, guest kernel, and trusted guest supervisor. Hardware-backed attestation is not established by the reported KVM integration [21].

A model identity hash records a supplied identity string; it does not prove which weights actually executed. An intent hash binds a recorded intent string; it does not prove internal goals. These distinctions prevent provenance metadata from being mistaken for hardware or semantic attestation [18].

C. Objectives and Exclusions

The objectives are to prevent ungranted execution, constrain admitted work, limit delegated authority, restrict broker-mediated effects, detect selected trusted-context leakage, and distinguish verified completion from denial or interruption. Mandatory production protections are intended to fail closed when they cannot be established.

Explicit exclusions include malicious administrators, compromised kernels/hypervisors, all hardware side channels, defects absent from the test and formal abstractions, and harmful actions that remain entirely within explicitly granted authority. BULL does not prove that permitted work is useful, correct, or benign [16].

IV. ARCHITECTURE AND CURRENT IMPLEMENTATION

BULL separates an authority-owning control plane from a workload plane that proposes or performs work. Production entry points are explicit `ProductionRuntime` and `ProductionDispatcher` objects. The compatibility dispatcher rejects the legacy attempt to establish a production boundary with only a Boolean flag, and the production wiring checks the actual configured runtime [17].

A. Canonicalization and Exact Execution Binding

A `DispatchRequest` separates the model-side proposal from trusted context and granted capabilities. When a security-domain registry is configured, trusted identity and parent authority come from host state rather than model-controlled metadata. Canonicalization maps the operation to a resource and capability before dispatch [17].

Command execution requires `process.exec`. The canonical executable must match the authorized resource. Production additionally requires a host-authorized complete argument

vector, and execution is rejected when the actual vector differs. Thus, authorization is bound to what is executed rather than to a broad natural-language task label [17].

Broker authorization has a distinct boundary. Egress and secret operations require an explicit `ALLOW`; a `SANDBOX` verdict cannot pass a host broker that lacks a corresponding sandbox mediation step. This exact bypass condition became a permanent regression test before the current evidence snapshot [15], [17].

B. Security Domains and Delegated Authority

The domain registry assigns host-generated identities with root and parent lineage, capability ceilings, approved egress hosts, provenance, intent and command hashes, optional model identity, and freeze state. Child capabilities and hosts are restricted to subsets of parent authority. Siblings retain distinct identities while sharing root behavioral security history, so spawning another child does not automatically reset the monitor [18].

These controls provide policy continuity and attribution; they are not a general proof that covert collaboration will always be detected. A shared external service that is legitimately authorized still requires its own policy analysis.

C. Multi-Agent Orchestration and Synthetic Canary Boundary

The current repository also contains a structured multi-agent orchestration layer. A coordinator normalizes tasks, a default-deny policy agent issues one-time approval tokens, an executor runs registered callables rather than `eval/exec`, and independent auditor/verifier stages scan outputs. A capability-checked message bus limits hops and records structured trace events [19].

A host-owned synthetic honeypot is minted per run. The raw canary is permitted only in one exact initial trusted-context field and message. The boundary guard inspects bus envelopes, agent results, and adapter-routed external writes for the raw marker and selected encodings or split-field reconstruction. A detected leak trips the trace, suppresses contaminated output, and records only a token fingerprint. Oversized uninspectable payloads fail closed with a scan-overflow finding [19].

The canary is a detection signal, not a proof of safety when absent. Arbitrary encryption, semantic transformation,

or action-only attacks can avoid reproducing the marker. It therefore complements deterministic authorization and isolation rather than replacing them.

D. Admitted Snapshots and Malware Scanning

The filesystem pipeline validates project content, copies it to isolated snapshot storage, validates the copy, scans it, and binds execution to the admitted content. Unsupported file structures and detected mutation fail closed instead of executing directly from a changing workspace [16], [21].

The integration work corrected cross-filesystem directory hashing and trusted snapshot-worker imports. It also increased the bounded ClamAV scanner ceiling to 1536 MiB after the full database could not load under the earlier 768 MiB ceiling; the separate workload address-space limit remained 512 MiB in the recorded guest experiment [21]. A successful malware scan is evidence for the configured scanner/database, not proof that unknown malicious behavior is absent.

E. Operating-System Enforcement and Live Bootstrap Evidence

The production profile combines Linux namespaces, `no_new_privs`, a default-deny seccomp allowlist, Landlock restrictions, resource limits, and a trusted read-only `/bull_runtime` mount distinct from the workload workspace [16]. Linux documentation itself distinguishes syscall filtering from a complete sandbox; the security claim therefore concerns the composed boundary rather than any single kernel mechanism [11], [12].

The runtime bootstrap returns nonce-bound evidence through a parent-created descriptor before workload execution. The reported guest configuration observed Landlock ABI 9, strict seccomp, isolated namespaces, and loopback-only networking. The observed strict syscall policy contained 170 rules [21]. These are measurements for the tested guest stack, not universal Linux properties.

F. Secrets, Egress, and Audit

Secret grants can be TTL/use-limited and domain-bound. Egress authority is constrained by both domain host authority and broker policy. Broker operations are re-authorized rather than treated as shortcuts around the reference monitor [16], [17].

Audit records are hash chained and can be checkpointed to an external service. The production HTTPS transport uses credential-free HTTPS endpoints, forbids redirects, sends a versioned authenticated checkpoint, requires HTTP 200, decodes the returned acknowledgement, and verifies that the acknowledgement matches the expected session, sequence, and record hash. The current transport explicitly sets `User-Agent: BULL-AuditAnchor/1` after an operator-run edge compatibility failure with the default Python client signature [25], [28].

G. One-Shot QEMU/KVM Integration

The MicroVM integration provisions a fresh VM and bounded authority record for one request. The tested launcher attached read-only root, workspace, and runtime images, exposed no guest NIC, and used dedicated authenticated control and audit channels. The guest supervisor validated immutable deployment authority, established the audit relay, initialized the production runtime, and accepted only the request bound to that authority. Workload output remained data rather than control [20]–[22].

The supervisor checked for remaining workload cgroups and snapshots before completion. The host launcher supervised and reaped the exact QEMU child. Persistent VM reuse remains deferred; the one-shot results cannot be transferred to a long-lived multi-request service without new revocation, state-isolation, and recovery analysis [21].

H. Production Host Provisioning and External Audit Deployment

The current main branch includes a one-command Linux x86-64 production provisioner. It requires a non-local HTTPS audit endpoint and an owner-only anchor key, obtains a clean source revision, installs the package, creates private state, establishes a delegated cgroup-v2 parent, generates signed integrity and policy bundles, creates a fresh audit session, runs `bull verify -production`, exercises a real remote checkpoint/acknowledgement, and constructs the production runtime/dispatcher. The script prints success only after these stages complete [27].

Main now also contains an optional Cloudflare Workers + D1 reference deployment for the audit anchor. Its documented contract stores checkpoint continuity in D1, accepts authenticated POST checkpoints on a dedicated route, permits only an exact retry of the latest committed frame, and returns a protocol-compatible authenticated acknowledgement [26]. Cloud-account compromise remains part of that external-service trust surface.

V. SECURITY CONTRACT AND BOUNDED FORMAL MODELS

Let a request be $r = (\text{operation}, \text{resource}, \text{argv}, \text{context})$. Let C_d be the capability ceiling of domain d , E_d its approved egress-host set, and $G(r)$ the mandatory deployment/admission gates. The intended contract can be summarized as

$$\text{Executed}(r) \implies \text{Authorized}(r) \wedge G(r),$$

$$\text{Executed}(r) \implies \text{BindingMatches}(r),$$

$$C_{\text{child}} \subseteq C_{\text{parent}},$$

$$E_{\text{child}} \subseteq E_{\text{parent}},$$

$$\text{TrustedSuccess}(r) \implies \text{AuthenticatedResult}(r),$$

$$\text{TrustedSuccess}(r) \implies \text{AcceptedAuditCheckpoint}(r).$$

Authorization is path-specific. A sandboxed execution decision is not interchangeable with an explicit broker `ALLOW`. $G(r)$ represents successful configured checks, not an oracle for

TABLE II
ARCHITECTURE COMPONENTS AND CURRENT EVIDENCE SOURCES

Layer	Role	Evidence used in this report
Dispatcher	Canonical resource, capability, exact executable and full argv binding	Source inspection and regressions [17], [31]
Security domains	Host identity, delegation ceilings, lineage and freeze state	Source and contract [18]
Multi-agent layer	Coordinator/policy/executor/auditor/verifier plus canary boundary	Documentation, package smoke test, red-team benchmark [19], [23]
Admission/sandbox	Snapshot, scan, namespaces, seccomp, Landlock, limits	Production docs and real-KVM report [16], [21]
Guest supervisor	One-shot authority, trusted result and cleanup handling	Source and five-case KVM report [20], [21]
Audit transport	Authenticated checkpoint/ack, fail-closed recovery	Source plus operator external-anchor evidence [25], [28]
Provisioning	Signed deployment state, cgroups, verification and runtime construction	Current provisioner [27]

all malicious behavior. A failed audit acknowledgement also cannot retroactively undo an external side effect that already occurred.

The runtime TLA+ configuration checks invariants including `DenyNeverExecutes`, `EscalateNeverExecutes`, `ExecutionRequiresApprovedDecision`, `ExecutionRequiresCleanScan`, `ExecutionAlwaysSandboxed`, `ExecutionAlwaysSeccomp`, `ChildAuthorityBound`, `SecretRequiresGrant`, `EgressRequiresBroker`, and `DirectNetworkAlwaysBlocked` [29]. The session/audit model checks `DurableBeforeAcknowledged`, `AcknowledgedBeforeExecuted`, `RemoteNeverAheadOfLocal`, `ConflictStopsExecution`, and `SequentialHistory`, with `MaxCheckpoints=3` in the checked configuration [30].

On the current main CI run, TLC reported no error after generating 1055 states and finding 405 distinct runtime states, and no error after generating 69 states and finding 57 distinct session/audit states; both queues ended empty [33]. These bounded state-space results make intended transitions explicit. They are not a refinement proof of Python, Linux, QEMU/KVM, Cloudflare, or hardware.

VI. EVALUATION AND EVIDENCE

A. Evidence Generations

The current evidence is intentionally split by revision and environment. The historical KVM integration used a source-tree hash recorded before its follow-up documentation commits. The operator red-team benchmark pinned commit `cd461ae`. The current hosted CI and publication snapshot use main commit `01cd259`. The external audit note records an operator-controlled production host and non-local collector. These results should not be arithmetically combined as if they were one monolithic test run.

B. Current Main Continuous Integration

Table III records the hosted checks for commit `01cd259`. The 32-test local-host red-team job is a focused subset executed before the full suite and should not be added to

the 272-test total. Likewise, the two Python-version runs are compatibility repetitions, not 544 unique security properties.

All four push workflows associated with the current main snapshot—security regression, MicroVM host regression, formal invariants, and Pages deployment—completed successfully [31]–[34].

C. Historical Real-KVM Integration

The project’s September 20 local integration report records five real QEMU/KVM cases, all passing with none skipped. That run is distinct from hosted CI and used a disposable authenticated local TLS collector rather than the later external production collector [21].

The same historical report recorded 217 passing host tests plus two subtests at the tested KVM source snapshot, Landlock ABI 9, strict seccomp, `no_new_privs`, read-only input/runtime mounts, a 512 MiB workload address-space limit, descriptors 0–2, authenticated session-bound control/audit, and cleanup checks. Malformed/missing/replayed protocol evidence was covered by host regressions rather than extra KVM cases. Hardware-backed attestation and exhaustive escape testing were not performed [21].

D. Single-Run KVM Timing Observation

The allowed KVM run reported the intervals in Table V. They overlap and must not be summed. These are one-run observations, not percentiles or a comparative performance study.

For this fixture, startup and admission dominate useful workload time. That result identifies an optimization target but does not justify weakening admission, syscall, or audit controls. A persistent VM could reduce cold-start cost while simultaneously increasing stale-state and cross-request risk.

E. Operator Red-Team Benchmark

The benchmark now published in the repository pins the tested source to commit `cd461ae` and records the host as Linux 7.2.6-1-cachyos, Python 3.14.7, an AMD Ryzen 5 7430U (12 logical CPUs), and 11 GiB memory. Eleven selected defensive regression classes were executed three times

TABLE III
HOSTED CI RESULTS ON MAIN COMMIT 01cd259

Workflow/job	Observed result	Timing/state count	Interpretation limit
Security regression, Python 3.11	272 passed; 15 warnings; 6 subtests passed	3.76 s; local-host subset 32 passed in 0.34 s	Warnings are expected development-API warnings; test count is not attack-space coverage.
Security regression, Python 3.13	272 passed; 15 warnings; 6 subtests passed	3.49 s; local-host subset 32 passed in 0.24 s	Compatibility repetition of the same repository corpus.
Installed-package smoke	import bulldog and bulldog.multiagent passed	Both Python versions	Importability is packaging evidence, not runtime containment.
MicroVM host regression, Python 3.11	39 passed	2.06 s	Host-side regression; not a KVM boot.
MicroVM host regression, Python 3.13	39 passed	2.15 s	Host-side regression; not a KVM boot.
TLA+ runtime	No error	1055 generated / 405 distinct states	Bounded abstraction only.
TLA+ session/audit	No error	69 generated / 57 distinct states	Bounded abstraction only.
Pages verify/deploy	Build and deployment passed	183 tracked regular files indexed	Publication integrity, not security certification.

TABLE IV
FIVE REPORTED REAL-KVM CASES

Case	Project-reported observation	What the observation establishes
Authorized request	ProductionDispatcher → ProductionRuntime → actual workload, exit status 0	The tested permitted path executed through the integrated guest stack.
Denied request	DENY, executed=false, no sandbox/workload execution	The tested denial stopped before workload execution.
Timeout	No successful completion; no remaining workload cgroups or snapshots	The tested deadline path did not masquerade as success and reported cleanup.
Cancellation	Exact QEMU child reaped; missing completion remained interrupted; no replay	The tested host cancellation did not create a false completed result.
Missing protection	ProductionGateFailure before request admission	The tested absent mandatory protection failed closed.

TABLE V
SINGLE ALLOWED-RUN KVM TIMINGS

Interval	ms
QEMU start → dispatcher-ready	12,157.320
Launcher → dispatcher-ready	12,282.303
Guest runtime/dispatcher init	597.739
Dispatch incl. admission & scan	9,165.705
Sandbox setup → pre-exec	308.901
Pre-exec → process reap	99.492
Deterministic fixture body	0.476
Completion audit acknowledgement	33.714
Result/audit overhead outside dispatch	34.355

each, producing 33 passes and zero failures. A clean full regression invocation also returned PASS in 4.587 s [23], [24].

The scenario latency values are end-to-end `pytest` measurements that include interpreter, framework, fixture, and process overhead. They must not be labeled pure policy-decision latency. A PASS means the regression behaved as asserted for that corpus; it is not independent certification or evidence about unknown attack classes [23].

F. Direct Hot-Path Measurements

The same benchmark package records the in-process measurements in Table VII. Traversal rejection is a tight primitive measurement. The prompt-injection and benign multi-agent loops intentionally reuse one `MultiAgentSystem`; trace and honeypot state therefore grow during the 500-iteration run. Those numbers characterize accumulated-state/long-session behavior, not fresh-instance request latency [24].

The large difference between traversal rejection and stateful multi-agent execution is expected from the different work performed and benchmark construction. The present data do not isolate individual coordinator, canary, policy, auditor, or verifier costs.

G. External Production Audit-Anchor Evidence

A later operator-run deployment exercised the production gate with strict seccomp, signed integrity and policy state, private snapshot/audit paths, delegated cgroup v2 state, a fresh audit session, and a non-local HTTPS transport. The published evidence note records `PRODUCTION BOUNDARY: PASS` [25].

TABLE VI
PUBLISHED OPERATOR RED-TEAM BENCHMARK AT COMMIT `cd461ae`

Scenario	Surface	Runs	Pass	Mean ms	Peak RSS MiB
Path traversal encodings	filesystem	3	100%	338.406	45.500
Read authority → shell execution	authorization	3	100%	366.427	44.883
Authorized argv → force-push drift	authorization	3	100%	318.111	44.926
External content → credential read	secrets	3	100%	301.671	44.855
SANDBOX verdict → egress broker	egress	3	100%	292.603	44.734
SANDBOX verdict → secret broker	secrets	3	100%	304.441	44.668
Private/metadata SSRF targets	egress	3	100%	355.926	44.797
Wrong Unix peer UID	secrets	3	100%	360.088	44.809
Prompt injection	multi-agent	3	100%	351.591	46.086
Forged policy verdict	multi-agent	3	100%	367.611	45.012
Approval-token replay	multi-agent	3	100%	376.640	44.992

TABLE VII
PUBLISHED HOT-PATH MEASUREMENTS

Benchmark	Iter.	μ s/op	ops/s
Traversal rejection	500	4.897	204,187.31
Prompt-injection denial	500	66,559.856	15.02
Benign multi-agent allow	500	438,559.052	2.28

The deployment also produced a useful negative result. Cloudflare rejected Python urllib’s default client signature before the Worker, and an earlier failed delivery left one local ledger record without the corresponding remote checkpoint. BULLrefused a subsequent append. After the explicit BULL-AuditAnchor/1 client identifier was added, the operator used `AuditLedger.reconcile_remote()`; local-only verification passed with one record, remote reconciliation passed, combined verification passed, a fresh authenticated append passed, and final verification passed with two records [25], [28].

This sequence is stronger evidence than a simple success-only demonstration because it exercises failure, refusal to advance, explicit recovery, and a fresh authenticated checkpoint. It remains a single operator-controlled deployment, not a third-party security audit. It does not establish resilience to Cloudflare-account compromise or to a compromised trusted host.

H. Earlier Production-Verification Failure

Before the external-anchor path was corrected, a local production-verification attempt recorded 257 passing tests, 13 failures, 15 warnings, and six passing subtests in 7.30 s while a live remote-audit environment was active. The failures were concentrated in audit tests whose writes were routed to an unavailable/misaddressed external endpoint, and a production gate attempt also exposed invalid session/key and delegated-cgroup configuration. This negative result is retained as engineering history rather than discarded. It led to stricter provisioning, explicit environment requirements, endpoint debugging, and the eventual authenticated recovery sequence described above. Because the earlier run used a different operational state, its 257/13 result should not be compared numerically with the clean current-main 272-pass CI corpus.

I. What the Evidence Does Not Measure

The current record does not provide a representative malicious-effect escape rate, false-denial rate over a broad benign workload distribution, adaptive-agent benchmark against the KVM path, multi-host latency distribution, long-running persistent-session availability, or a head-to-head comparison with other isolation systems. It also does not establish resistance to kernel/hypervisor compromise, external account takeover, hardware attacks, or harmful behavior entirely inside granted authority.

VII. EVALUATION PLAN FOR A SUBSEQUENT EMPIRICAL STUDY

A comparative study should hold workload intent, data, hardware, and authorization scope constant while varying enforcement layers. Candidate configurations include an isolated laboratory baseline, authorization alone, authorization plus host process isolation, and the complete one-shot MicroVM configuration. Baselines lacking protection should use synthetic disposable infrastructure rather than third-party targets.

Two outcome families are required. First, **unauthorized-effect rate** measures prohibited effects that actually occur among attempted prohibited actions. Second, **legitimate-task completion rate** measures authorized tasks that complete correctly. A system that denies everything may score well on the first measure while being unusable. Review outcomes, startup failures, timeouts, and crashes should be reported separately.

The workload set should include ordinary file processing, constrained code execution, delegated work, sensitive-data handling with synthetic records, approved network operations against controlled endpoints, and failure recovery. Data-disclosure tests should distinguish unauthorized destinations from unauthorized payloads sent to approved destinations.

Performance experiments should separate cold-start from reused-state conditions, pin complete hardware/software identities, and repeat each condition enough to characterize variance. Median and 95th-percentile latency, peak memory, CPU time, scan duration, audit delay, and cleanup latency should be reported with sample size and uncertainty. The current hot-path benchmark already demonstrates why fresh-state versus accumulated-state labeling matters.

AgentDojo provides a published environment for evaluating tool-using agents and prompt-injection defenses [9]. Connect-

ing BULL to such a benchmark would require an explicit mapping from benchmark actions to BULL capabilities, resources, and brokers; no such benchmark result is claimed here.

VIII. RELATED WORK AND POSITIONING

Protective layers around models. CaMeL separates control and data flows and applies capability-based policies to constrain tool use [8]. It is directly relevant prior work and prevents a claim that external model protection or capability-governed actions originated with BULL. BULL’s emphasis is the execution lifecycle and its integration with OS and VM boundaries, brokers, delegation state, and audit recovery.

Isolation systems. gVisor interposes an application kernel to reduce direct host-system exposure [13]. Firecracker studies lightweight virtualization for serverless workloads [14]. BULL’s reported guest implementation uses QEMU/KVM, not Firecracker or gVisor, and therefore does not inherit their measurements or guarantees.

Kernel mechanisms. Seccomp and Landlock are established Linux enforcement mechanisms with different scopes [11], [12]. BULL composes them with application-level authorization, admission, and audit; it does not claim to have invented them.

Risk frameworks. OWASP’s Top 10 for Agentic Applications 2026 provides a useful vocabulary for agent risks [10]. A threat-framework citation is not compliance certification and does not prove that every listed risk has been mitigated.

IX. DISCUSSION AND LIMITATIONS

The evidence now spans four distinct layers. First, current hosted CI shows that the integrated repository passes its Python, MicroVM-host, formal, packaging, site-build, and deployment checks on the pinned main revision. Second, the historical real-KVM experiment demonstrates a bounded production-dispatch path inside a hardware-virtualized guest. Third, the published operator benchmark exercises selected authorization, egress, secret, filesystem, and multi-agent regressions repeatedly and provides direct throughput measurements. Fourth, the production-audit experiment exercises a non-local authenticated checkpoint path, including an interrupted checkpoint and explicit recovery.

This broader evidence changes the status of some claims from the previous manuscript revision. The multi-agent package is now on main rather than an unmerged branch. The KVM integration is on main rather than only an integration branch. The repository now publishes machine-readable benchmark data and visualizations. A production host provisioner and Cloudflare Workers+D1 reference anchor are present. The source license is GPL-2.0-or-later [15], [35]. These are implementation/reproducibility advances, not an independent validation of security effectiveness.

The largest unresolved issue remains generalization. An approved command may still be harmful; an allowed service may receive an inappropriate payload; a trusted key may be mishandled; an external service account may be compromised; a kernel defect may undermine local evidence. Hashing does

not prove semantics, a malware scan does not prove benignness, and an audit acknowledgement does not roll back an already executed external side effect.

Persistent operation remains future work. Keeping a guest alive could reduce the roughly 12 s observed cold-start path, but it creates longer-lived credentials, state, and revocation obligations. Writable workspace promotion and artifact export also remain future scope in the recorded MicroVM milestone. Hosted CI still performs MicroVM host regressions rather than a required real-KVM boot on every change [21].

Independent assessment would materially strengthen the project. The present results are authored, operated, and interpreted by the project owner. The appropriate claim is therefore that the implementation and evidence are increasingly inspectable and reproducible—not that the system is vulnerability-free or certified safe.

X. CONCLUSION

BULL was motivated by a practical question raised by recent AI security incidents: how can a system owner retain control over data and processes when an agent, its inputs, or an actor influencing it cannot be trusted to remain inside the assigned task? The architecture answers by treating model-generated actions as proposals, keeping authority in host-controlled state, binding authorization to actual operations, constraining delegated agents, isolating admitted workloads, brokering sensitive effects, and requiring trusted completion evidence.

The updated evidence snapshot is materially stronger than the earlier manuscript. Current main passes a 272-test regression corpus plus six subtests on two Python versions, 39 MicroVM-host regressions on each version, two bounded TLA+ models, and the publication pipeline. Historical real-KVM integration passed five of five cases. A published red-team benchmark recorded 33 of 33 selected runs passing. A production-audit deployment exercised fail-closed interruption, reconciliation, and a fresh authenticated checkpoint. None of these results alone or together proves universal safety. They do support continued empirical investigation of a concrete infrastructure-level approach: unauthorized consequences should be constrained by owner-controlled enforcement rather than left solely to the model’s interpretation of what it ought to do.

DECLARATIONS

Research and authorship. This manuscript describes the BULL project under Luis Minier / Anharmonyclabs. The author is a project stakeholder; the document is not an independent assessment.

AI assistance. ChatGPT assisted with manuscript drafting, source discovery, repository inspection, and synthesis of project evidence. AI-generated text is not experimental evidence. Test counts and measurements are attributed to CI logs, repository evidence files, or operator-run records.

Responsible reporting. Public incidents are discussed only at the level needed to motivate defensive controls. This report

does not publish deployment secrets, private keys, or sensitive workload data. Named third parties have not endorsed BULLthrough citation.

APPENDIX A VERSION AND EVIDENCE IDENTITY

A1. Current public manuscript baseline. Repository: Anharmoniclabs/BULL; branch: main; full commit identity is available through the pinned reference [15]. This signed commit publishes benchmark data/visualizations, operator external-audit evidence, and the Cloudflare Workers+D1 reference deployment package [15].

A2. Operator benchmark identity. Source commit cd461ae; UTC run 20260920T170536Z; Linux 7.2.6-1-cachyos x86-64; Python 3.14.7; AMD Ryzen 5 7430U; 12 logical CPUs; 11 GiB memory. The complete commit is stored in the machine-readable benchmark record [24]. The current repository embeds the corresponding machine-readable JSON, attack-summary CSV, hot-path CSV, and host metadata [24].

A3. Historical real-KVM identity. Tested base commit begins d218e688; the report records source-tree SHA-256 beginning 6316b6ea and the allowed-session identifier beginning 8ffa547a. Full values are preserved in the pinned integration report [21]. The report states that all five final KVM cases used the recorded source-tree identity; raw case evidence and deployment assets were retained outside Git [21].

A4. Current CI run identities.

Security regression:	35525606273
MicroVM host regression:	35525606201
Formal invariants:	35525606295
Pages verify/deploy:	35525606425

Each run is associated with main commit 01cd259 and completed successfully [31]–[34].

APPENDIX B CLAIM-TO-EVIDENCE LEDGER

APPENDIX C CURRENT KNOWN LIMITS AND NEXT EXPERIMENTS

- 1) Repeat the real-KVM suite on the exact current main revision rather than transferring historical KVM evidence forward by assumption.
- 2) Add a required hosted/self-hosted real-KVM CI lane where suitable hardware and security policy allow it; retain the existing host-only MicroVM regression lane separately.
- 3) Measure fresh-instance and long-session multi-agent throughput separately, including state-size growth and cleanup behavior.
- 4) Evaluate false denials and legitimate-task completion alongside blocked malicious effects.
- 5) Exercise the external audit anchor under collector loss, delayed acknowledgements, retry, key rotation, and account-level failure conditions.
- 6) Evaluate persistent MicroVM reuse only after defining per-request authority, revocation, credential lifetime, state separation, and no-replay recovery behavior.

- 7) Obtain independent reproduction and security review before making certification-style claims.

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TABLE VIII
CLAIMS, SUPPORTING EVIDENCE, AND BOUNDARIES OF INTERPRETATION

Claim	Evidence	Not established
Production dispatch binds full command	Dispatcher source and authorization regressions [17], [31]	That every authorized executable is safe.
Domain state limits delegated authority	Registry contract and multi-agent tests [18], [19]	Verified model weights or universal goal-drift detection.
Real guest production path executed	Five-case KVM report and guest supervisor source [20], [21]	Independent reproduction or exhaustive escape resistance.
Current integrated suite is green	272 tests + 6 subtests on both Python versions; all current push workflows successful [31]–[34]	Complete attack-space coverage.
Selected red-team corpus repeated cleanly	33/33 operator runs passed at pinned commit [23]	Unknown attacks or independent certification.
Traversal primitive is fast on recorded host	4.897 μ s/op, 204,187.31 ops/s [24]	End-to-end agent-task latency.
External checkpoint recovery worked once	Published operator evidence: reconcile, fresh append, authenticated ack, final valid ledger [25]	External-account compromise resistance or general availability.
Formal abstractions satisfy checked invariants	405 and 57 distinct-state model checks completed with no error [33]	Formal proof of the full Python/Linux/QEMU/cloud stack.

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